

Ph.D. Preliminary Examination (Analysis)

August, 2013

INSTRUCTIONS: *Do all six problems. In order to receive maximum credit, solutions to problems must be clearly and carefully presented and should contain the necessary details. All problems are worth the same number of points.*

1. Let X be a nonempty set, and for any two functions $f, g \in \mathbb{R}^X := \{h : h \text{ is a function from } X \text{ to } \mathbb{R}\}$, let

$$d(f, g) = \sup_{x \in X} \frac{|f(x) - g(x)|}{1 + |f(x) - g(x)|}.$$

Prove that $(\mathbb{R}^X, d)$ is a metric space.

2. Prove that $\sum_{n=1}^{\infty} \frac{nx^n}{x^n + 1}$ is not uniformly convergent on $[0, 1)$, but it defines a continuous function on $[0, 1)$.

3. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be twice continuously differentiable, $a, b \in \mathbb{R}$ and $a < b$. Suppose that $f(a) = f(b) = 0$ and $|f''(x)| \leq 1$ for every $x \in [a, b]$. Prove that

$$|f(x)| \leq \frac{(b-a)^2}{8}$$

for every $x \in [a, b]$.

4. Let $f(x) = f(x_1, x_2, x_3, x_4)$ be a C^1 function from $\mathbb{R}^4$ to $\mathbb{R}$ such that $f(0) = 0$, where 0 denotes the origin in $\mathbb{R}^4$. Suppose that

$$(\partial f(0)/\partial x_3, \partial f(0)/\partial x_4) \neq (0, 0).$$

Prove that there exist an open neighborhood U of $(0, 0)$ in $\mathbb{R}^2$ and two C^1 functions $\phi, \psi : U \rightarrow \mathbb{R}$ such that

$$f(s, t, \phi(s, t), \psi(s, t)) = 0 = f(s, t, -\psi(s, t), \phi(s, t))$$

holds for every $(s, t) \in U$.

5. Let $\Omega_1 = [0, 2] \times [0, 1]$ and $\Omega_2 = [1, 3] \times [0, 1]$. For each $j \in \{1, 2\}$, let $f_j : \Omega_j \rightarrow \mathbb{R}$ be Riemann integrable over Ω_j . Define $F : [0, 3] \times [0, 1] \rightarrow \mathbb{R}$ by

$$F(x, y) = \begin{cases} f_1(x, y) & \text{if } (x, y) \in \Omega_1 \setminus \Omega_2; \\ f_2(x, y) & \text{if } (x, y) \in \Omega_2 \setminus \Omega_1; \\ \inf\{f_1(x, y), f_2(x, y)\} & \text{if } (x, y) \in \Omega_1 \cap \Omega_2. \end{cases}$$

Prove that F is Riemann integrable over $[0, 3] \times [0, 1]$.

6. Let $\Omega = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 \leq 1\}$ and $f \in C^\infty(\mathbb{R}^3)$. Suppose that $f(x, y, z) = 0$ for all $(x, y, z) \in \partial\Omega$. Prove that

$$\left| \iiint_{\Omega} f(x, y, z) dV \right| \leq \frac{2\sqrt{5}\pi}{15} \left(\iiint_{\Omega} |\nabla f(x, y, z)|^2 dV \right)^{1/2}.$$